

The Thue-Morse Sequence

Fall 2025 ARML Power Contest

The Background

Mathematics abounds in curious sequences, and the Thue-Morse sequence is definitely among them. These questions will derive the sequence in a number of ways and demonstrate a few of its nifty properties and applications.

In this set of questions, a *word* is any ordered list of symbols taken from the set $\{0, 1\}$. Note that order of symbols matters! The “empty word” consisting of no symbols is allowed, as are infinitely long strings of symbols. Thus, 010100, 111, 0, and 101010... are all words. So is the empty word consisting of no symbols. The words 10 and 01 are different because the symbols are in different orders. If W is a word of finite length, and X is any word, then WX is simply the word consisting of the symbols in W followed by those in X . The word X is a *subword* of W if $W = YXZ$ where Y and Z are any words (including the empty word). Thus 01 and 10 are both subwords of 101, but 11 is *not* a subword. The *length* of a word is the total number of symbols in it, including repeats. Thus the length of 10110 is 5.

The Problems

There are 40 points available on this contest. The numbers at the end of each problem are the number of points that problem is worth. These are also indicated on the cover sheet, which should be the front page of your solution packet. Please make sure to write on only one side of the answer sheets.

1. List all positive integers less than 100 that are perfect powers. (That is, numbers n such that $n = a^b$ where a is an integer and b is a positive integer greater than 1.) [2 pts.]
2. Among these perfect powers, several combinations add to zero, where a *combination* is any sum or difference of distinct numbers. For instance, $9 + 16 - 25 = 0$ or $1 + 8 + 9 - 16 + 25 - 27 = 0$. List three more such combinations, each using at most six of the numbers in the list from problem 1. (changing all the signs in a combination does not count as a separate combination!). [2 pts.]
3. One of the most interesting combinations that adds to zero is this string of consecutive squares:

$$1^2 - 2^2 - 3^2 + 4^2 - 5^2 + 6^2 + 7^2 - 8^2 = 0.$$

In fact, prove that for *any* 8 consecutive integers (not necessarily positive) $a, a + 1, \dots, a + 7$, using the same signs as in the above expression, [2 pts.]

$$a^2 - (a + 1)^2 - (a + 2)^2 + (a + 3)^2 - (a + 4)^2 + (a + 5)^2 + (a + 6)^2 - (a + 7)^2 = 0.$$

The pattern in this last question is no accident, and the rest of the questions in this contest explore it further.

Define a sequence of words as follows:

$$W_n = \begin{cases} 0 & n = 0 \\ W_{n-1} \overline{W_{n-1}} & n > 0 \end{cases},$$

where $\overline{W}$ means to take the *complement* of word W , by replacing each 1 in W with 0 and each 0 with 1. So $W_0 = 0$, $W_1 = 0\overline{0} = 01$, $W_2 = 01\overline{01} = 0110$, and $W_3 = 0110\overline{0110} = 01101001$.

4. Compute W_4 . [1 pt.]

5. Prove that the length of W_n is 2^n . [2 pts.]

Define s_k for $k \geq 0$ to be the $(k+1)$ -st letter counting from the left end of the word W_n , for some n so that $2^n > k$. For example, letting $n = 3$ allows $2^n = 8$ and looking at the first, third, and seventh letter in W_3 gives $s_0 = 0$, $s_2 = 1$ and $s_6 = 0$. Note that there is no actual dependence on n because once $2^n > k$, to create W_{m+1} from W_m for any $m \geq n$, all new letters are added to the right end of the word and the first 2^m letters remain the same.

The *Thue-Morse sequence* is the sequence $s_0, s_1, s_2, \dots = 0, 1, 1, 0, 1, 0, 0, 1, \dots$. You may think of it as the sequence of letters in the “infinite word W_∞ .”

6. Compute s_{23} . [1 pt.]

7. Prove W_n is a palindrome (that is, the word is left unchanged by reversing the order of its letters) if and only if n is even. [3 pts.]

8. Show that for any nonnegative integer, k , that the number of ones in the binary expression for k is even if $s_k = 0$ and odd if $s_k = 1$. For instance, $6 = 110_{(2)}$ has an even number of ones, and $s_6 = 0$. [3 pts.]

9. Let X be any word in the alphabet $\{0, 1\}$ and define a new word $\mu(X)$ by replacing each 0 of X by 01 and replacing each 1 with 10. For instance, $\mu(100) = 100101$. Prove that $\mu(W_n) = W_{n+1}$ and that $\mu(\overline{W_n}) = \overline{W_{n+1}}$. [3 pts.]

10. Prove that for any nonnegative integer k , $s_{2k} = s_k$ and $s_{2k+1} = 1 - s_k$. [2 pts.]

Let $p(x)$ be a polynomial. The degree of $p(x)$ will be denoted d and, as usual, is the highest power of x that occurs in the polynomial $p(x)$. Let a be any real number. Given $p(x)$, define $\Delta_a p(x) = p(x) - p(x+a)$. For instance, for $p(x) = 2x^3 - x^2 + x + 6$, $p(x+4) = 2(x+4)^3 - (x+4)^2 + (x+4) + 6 = 2x^3 + 23x^2 + 89x + 122$ and $\Delta_4 p(x) = -24x^2 - 88x - 116$.

11. (a) Let $p(x)$ be any constant polynomial, $p(x) = c_0$, and let a be any real number. Show $\Delta_a p(x) = 0$. [1 pt.]

(b) Let $p(x) = c_d x^d + c_{d-1} x^{d-1} + \dots + c_1 x + c_0$ with $c_d \neq 0$ be any polynomial of positive degree, $d > 0$, and let a be any nonzero real number. Show that $\Delta_a p(x)$ has degree exactly $d - 1$. [3 pts.]

(c) Let $p(x)$ have degree d and let $a_1, a_2, \dots, a_d, a_{d+1}$ be any $d+1$ real numbers. Show that $\Delta_{a_1}(\Delta_{a_2} \cdots \Delta_{a_d}(\Delta_{a_{d+1}} p(x)) \cdots) = 0$. [3 pts.]

Let $p(x)$ be any polynomial. For nonnegative integer k , define the polynomial $p_k(x)$ as follows:

$$p_k(x) = \begin{cases} p(x) & k = 0 \\ \Delta_{2^{k-1}} p_{k-1}(x) & k > 0 \end{cases}.$$

So $p_0(x) = p(x)$, $p_1(x) = \Delta_{2^0} p_0(x) = \Delta_1 p(x) = p(x) - p(x+1)$ and $p_2(x) = \Delta_{2^1} p_1(x) = \Delta_2(p(x) - p(x+1))$ which simplifies to $p(x) - p(x+1) - p(x+2) + p(x+3)$.

12. For any polynomial $p(x)$ and any nonnegative integer n , prove that

$$p_n(x) = \sum_{j=0}^{2^n-1} (-1)^{s_j} p(x+j)$$

where the s_j are the entries of the Thue-Morse sequence defined previously. [4 pts.]

13. Prove that for any nonnegative integer n , the combination of 2^{n+1} consecutive powers, using signs determined by terms of the Thue-Morse sequence ($0 \leftrightarrow +$, $1 \leftrightarrow -$), sums to zero, for any power up to and including n . (For example, with $n = 2$ the statement is that each of

$$\begin{aligned} a^0 - (a+1)^0 - (a+2)^0 + (a+3)^0 - (a+4)^0 + (a+5)^0 + (a+6)^0 - (a+7)^0 &= 0 \\ a^1 - (a+1)^1 - (a+2)^1 + (a+3)^1 - (a+4)^1 + (a+5)^1 + (a+6)^1 - (a+7)^1 &= 0 \\ a^2 - (a+1)^2 - (a+2)^2 + (a+3)^2 - (a+4)^2 + (a+5)^2 + (a+6)^2 - (a+7)^2 &= 0 \end{aligned}$$

for any a . Compare with problem 3.) [3 pts.]

14. Given 16 cubes whose edges have lengths 1, 2, ..., 16, separate the cubes into two subsets so that the number of cubes in each subset, the total length of all edges of all cubes in each subset is equal, the total surface area of all cubes in each subset is equal, and the total volume of the cubes in each subset is equal, simultaneously!. [1 pt.]
15. (a) Prove that neither word 000 nor 111 can appear as a subword of W_n . [2 pts.]
 (b) Prove that neither word 01010 nor 10101 can appear as a subword of W_n . [2 pts.]

Much more is true about the Thue-Morse sequence. Extending the last problem, the sequence is *overlap free*, meaning that no subword of the form $xWxWx$ can appear where x is any single letter and W is any word (possibly empty). The Thue-Morse sequence shows up in certain move sequences in chess, and has even been used to compose music! It is well worth poking around article on *Wikipedia* and *MathWorld* to start to learn more about this fascinating sequence.